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Oefeningenreeks 2E nr. 5.2

$$\begin{aligned} & \lim_{x \rightarrow 0} (1-x)^{\frac{a}{x}} \\ & \left(x = \frac{1}{t} \right) \\ & = \lim_{t \rightarrow \infty} \left(1 - \frac{1}{t} \right)^{at} \\ & = \lim_{t \rightarrow \infty} \left[\left(1 - \frac{1}{t} \right)^t \right]^a \\ & = \left(\frac{1}{e} \right)^a \\ & = e^{-a} \end{aligned}$$

References